

Towers of Hanoi Problem

The Tower of Hanoi problem was Invented by the French Mathematician Edouard Lucas in 1883.

It is a puzzle involving usage of stacks.

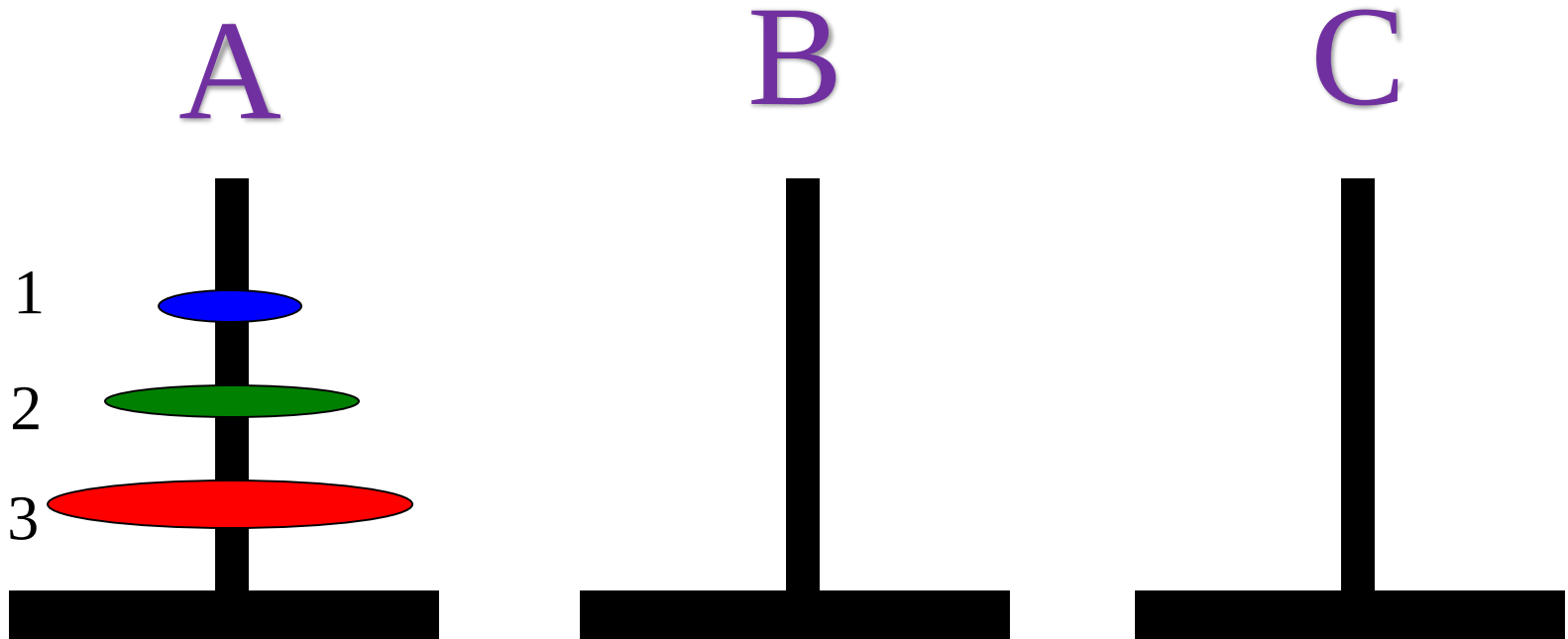
Task: Shifting of discs from source tower to destination tower.

Rules for solving puzzle:

1. One disc can be shifted at a time.
2. Only top disc can be shifted.
3. Large disc cannot be shifted on smaller disc.
4. Discs should be shifted in minimum number of moves

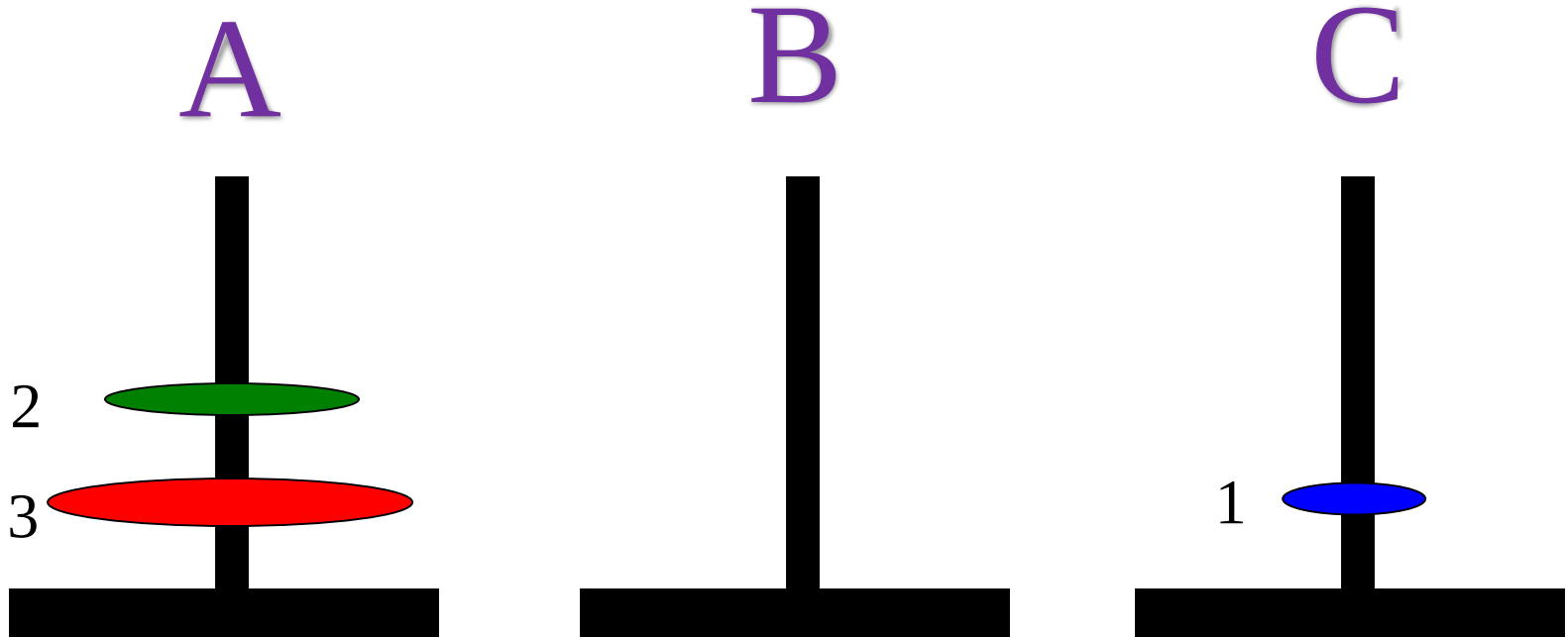
Source Tower having N discs can be shifted to destination tower using $(2^N)-1$ moves.

Tower of Hanoi



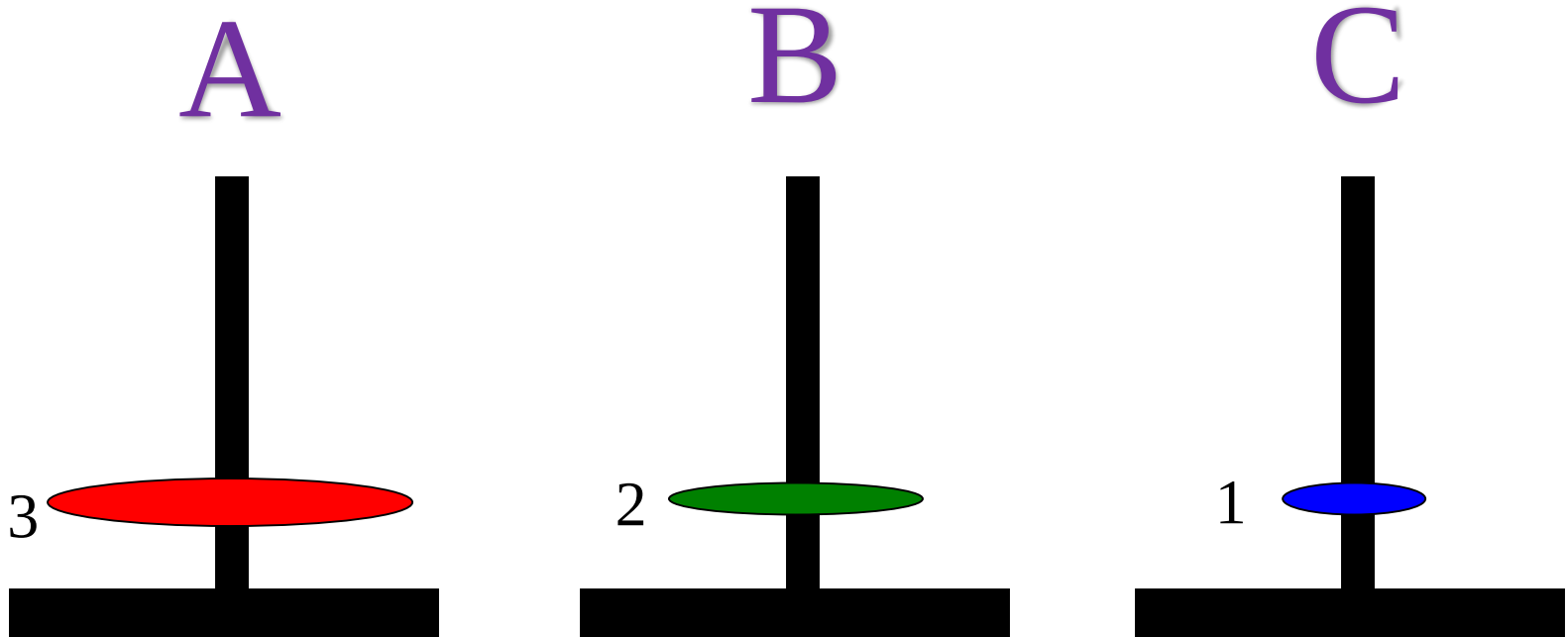
Tower of Hanoi

Move 1 from A to C



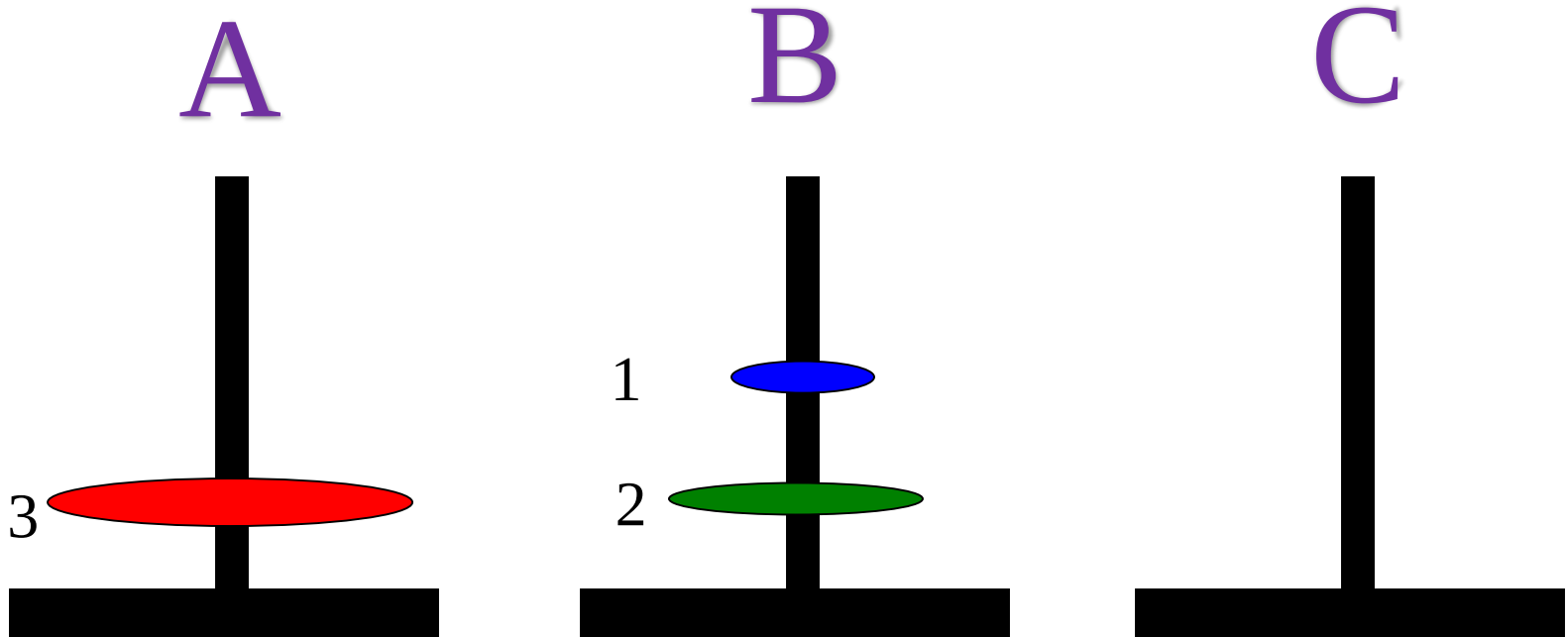
Tower of Hanoi

Move 2 from A to B



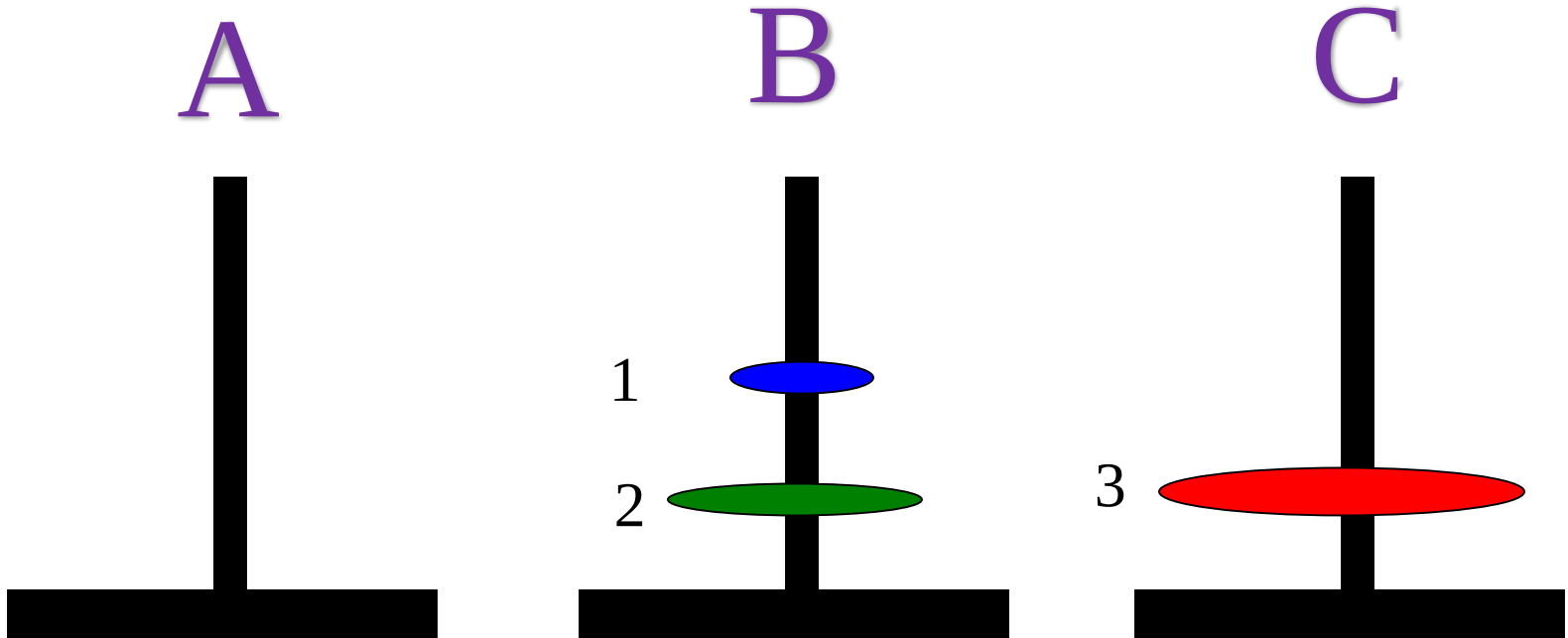
Tower of Hanoi

Move 1 from C to B



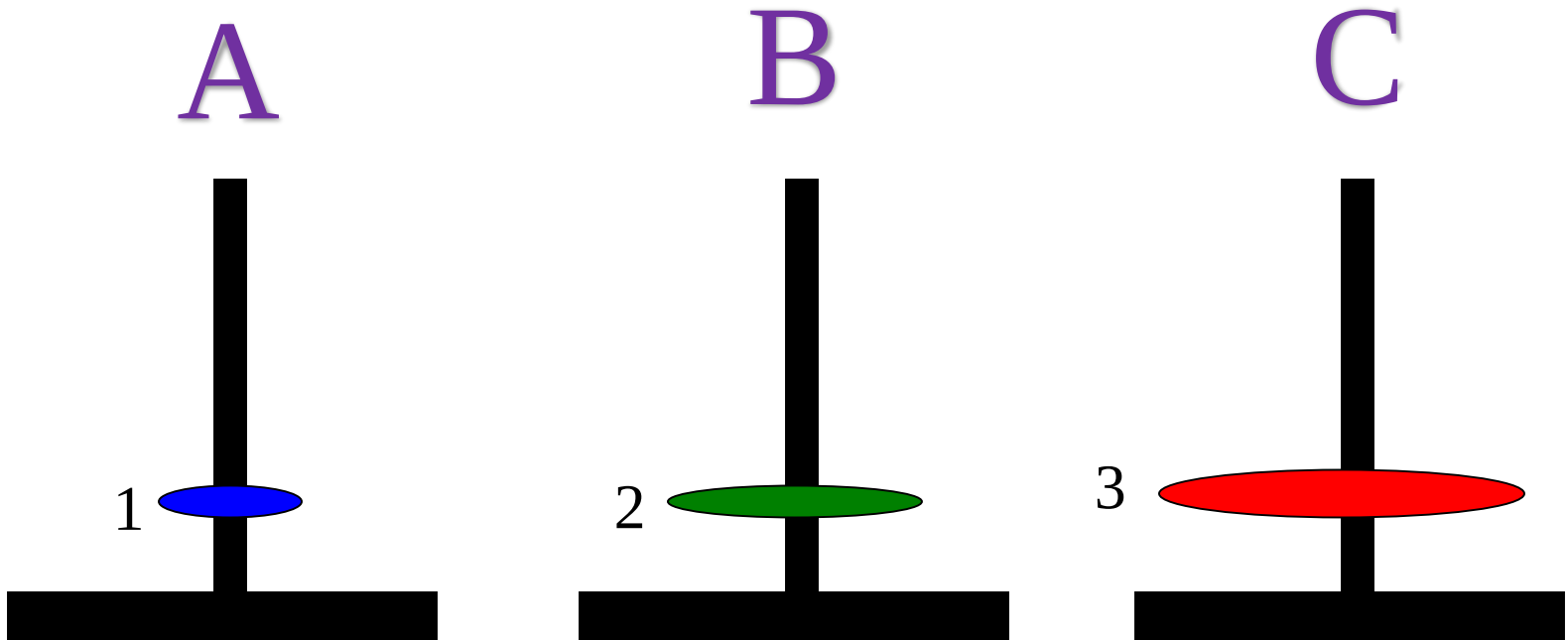
Tower of Hanoi

Move 3 from A to C



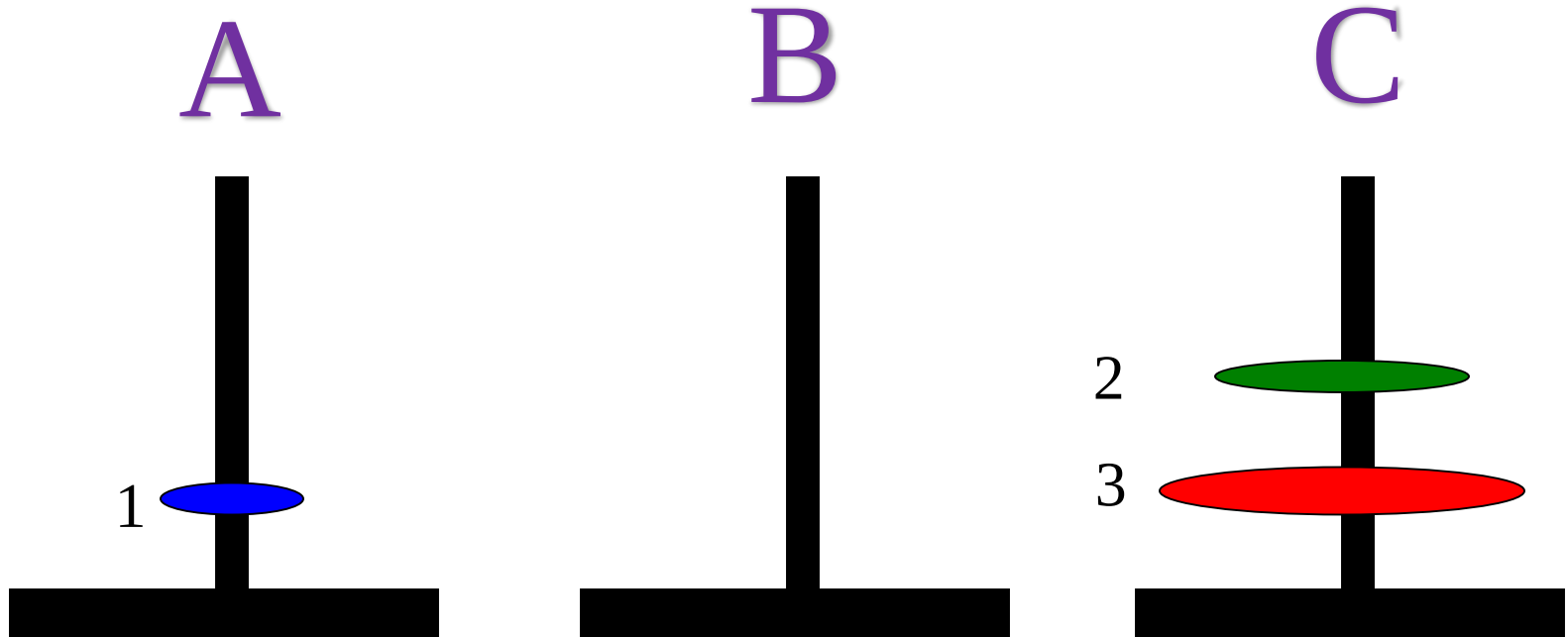
Tower of Hanoi

Move 1 from B to A



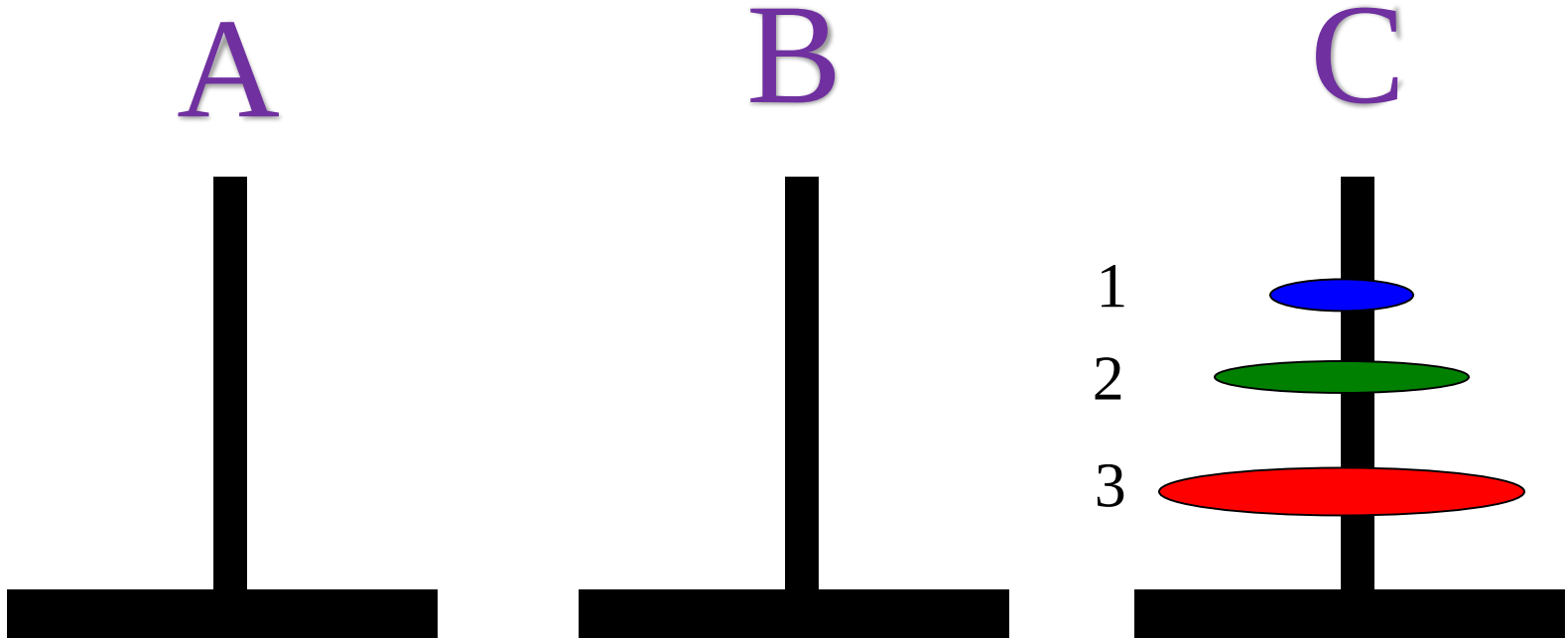
Tower of Hanoi

Move 2 from B to C



Tower of Hanoi

Move 1 from A to C



Recursive Algorithm

```
ToH(N, from_tower, to_tower, aux_tower)
{
    IF (N >=1) THEN
        ToH(N-1, from_tower, aux_tower, to_tower)
        print("Moving disc from", from_tower, "to", to_tower)
        ToH(N-1, aux_tower, to_tower, from_tower)
    END IF
}
```

Eg. ToH(3, 'A', 'C', 'B')

Applications

- Used as backup rotation scheme.
- Used by neuropsychologists trying to evaluate frontal lobe deficits.
- Used to solve mathematical problems related to Hamilton Cycle.

Analysis

- Let $T(n)$ is the time complexity for moving n discs.

$$T(n) = \begin{cases} 2T(n-1) + 1, & n > 1 \\ 1, & n = 1 \end{cases}$$